Software Development: Hospital Project

Testing

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Testing Plan

Test Strategy and Methodologies

- **Unit Testing:** Testing individual methods (e.g., validating that a "Patient" object is created correctly).
- **Integration Testing:** Ensuring the Java application communicates correctly with the Hospital database to save and retrieve data.
- **System Testing:** Testing the complete application from the perspective of a user following the User Guide.
- **Error Handling (Black Box Testing):** Verifying that the program appropriately handles invalid inputs with exceptions or pre-validation (e.g., preventing string input in numeric field)

Test Plan Structure

- **Reliability Testing:** Scenarios to ensure the system doesn't crash. For example, testing what happens if a user enters a string when an integer ID is required.
- **Performance Testing:** Assessing how the system handles the "hard limit" on records shown (e.g., viewing 100+ visits) to ensure it remains responsive.
- **Database Integrity:** Verifying that creating, updating, and deleting records in Java correctly reflects in the database.

Test Types

White Box Testing (Structural Testing)

This method focuses on the internal code logic, branches, and loops rather than just the user inputs.

- **Unit Testing:** Testing individual methods within the classes (e.g., verifying a specific calculation in the Prescription class) to ensure they work in isolation.
- **Statement Coverage:** Ensuring every line of the Java code is executed at least once during the testing process to find hidden bugs.
- **Path Testing:** Testing all possible paths through a switch or if-else block, such as the different branches in the navigation menus.

Gray Box Testing

This is a combination of both Black and White box techniques. Use knowledge of the database structure to test the application's functionality.

- **Integration Testing:** Verifying that the Java application correctly interacts with the Hospital database. Look at the database tables to see if a record added in the app actually appeared in the SQL table.
- **Regression Testing:** Re-running previous tests after a bug is fixed (like the InputMismatchException mentioned in the test log) to ensure the fix didn't break other parts of the system.

Non-Functional Testing

The marking scheme explicitly requires coverage of **Reliability and Performance**.

- **Reliability Testing:** Stressing the system by entering invalid data types or extreme values to ensure the "program appropriately handles errors with exceptions" without crashing.
- **Performance (Load) Testing:** Testing how the system handles the "hard limit" on records shown. For example, might test how quickly the app displays 50 visits versus 5 visits to evaluate data retrieval speed.

Test Log

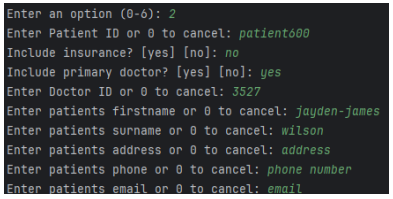
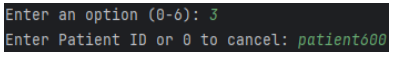
Test case	Test data	Expected result	Actual result	Comments	State

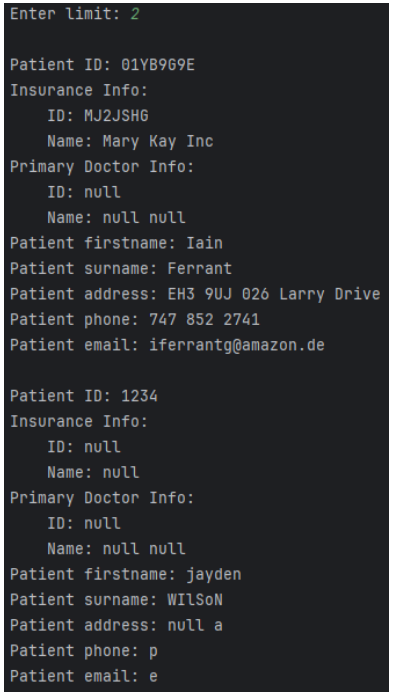
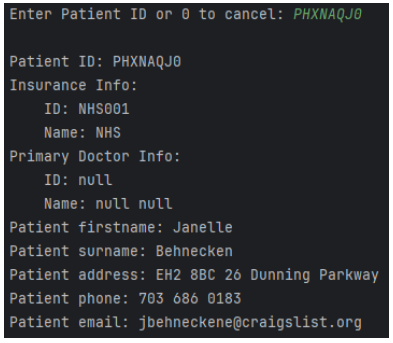
Early cancellation (global)	Patient ID: 0
	Insurance ID: 0
	Doctor ID: 0
	Doctors firstname: 0
	Insurance company name: 0
	patients address: 0
	Drug name: 0
		When 0 is entered the system should cancel the operation early	When 0 is entered the system cancel the operation early	`Enter Doctor ID or 0 to cancel: 0`
				`Operation cancelled`
				Same for others
					pass

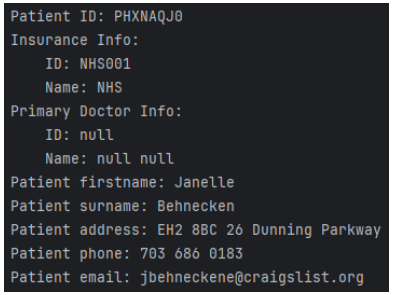
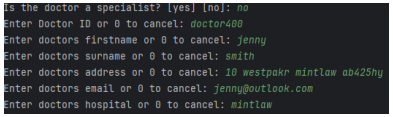
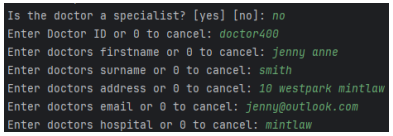
Global Negative Value Handling (All ID and fields across Doctors, Patients, Insurance, etc.)	-1, @#$%, NULL	System displays a "Validation Error" or "Invalid Input" message; system does NOT crash or save the record.	System displays error	`Enter Patient ID or 0 to cancel: -1` `Invalid input.`	pass

Date formatting	YYYY-MM-DD	System should only allow dates with the correct yyyy-mm-dd format	System only allows dates with the correct yyyy-mm-dd format	`Enter prescription date (yyyy-mm-dd) or 0 to exit: date`
				`Invalid date format. Please try again using YYYY-MM-DD.`
				`Enter prescription date (yyyy-mm-dd) or 0 to exit: 2025-04-90`
				`Invalid date format. Please try again using YYYY-MM-DD.`
					pass

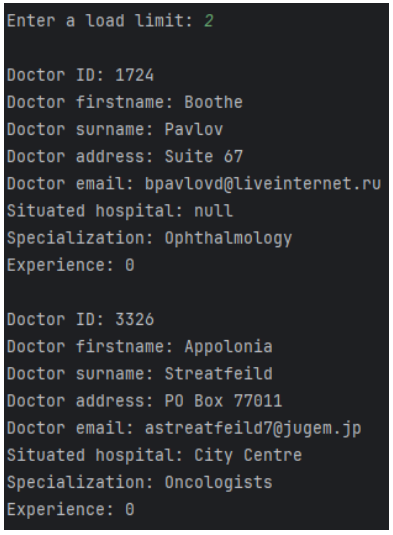
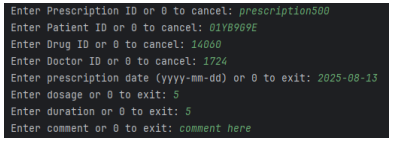
Performance (Retrieve 100 records using "Limit) for all modules	Limit: 100	Display within < 2 seconds	Display within < 2 seconds		pass
(Yes/No) Inputs (valid)	Include insurance?: no, n Include doctor?: no, n	When “no” is entered it should skip asking for that specific detail	Does in fact skip when no, n are entered	Include insurance? [yes] [no]: *no* Include primary doctor? [yes] [no]: *no*	pass
(Yes/No) Inputs (invalid)	Include insurance?: maybe Include doctor?: maybe	When “maybe” is entered it should ask for input again	Does ask for input again	Include insurance? [yes] [no]: *maybe* Invalid option. Please try again.	pass
Identifiers (Global)	Doctor Id: 0765 Insurance Id: 234567 Patient Id: 34567 Drug Id: 7129aba82 Visit id: 928272 Prescription id: 98765	When an id is entered that doesn't match an existing record it asks for id again unless “0” is entered for early cancellation	Asks for an id until a corresponding record is found	Enter Insurance ID or 0 to cancel: *234567* Insurance not found. Try again. Works the same for the rest	pass
Add Patient	Patient ID: patient600 Include insurance? [yes] [no]: yes Insurance ID: MJ2JSHG Include primary doctor? [yes] [no]: yes Doctor ID: 3527	Should add patent to database	Adds patient to database	Enter an option (0-6): *1* Enter Patient ID or 0 to cancel: *patient600* Include insurance? [yes] [no]: *yes* Enter Insurance ID or 0 to cancel: *MJ2JSHG* Include primary doctor? [yes] [no]: *yes* Enter Doctor ID or 0 to cancel: *3527* Enter patients firstname or 0 to cancel: *jayden* Enter patients surname or 0 to cancel: *wilson* Enter patients address or 0 to cancel: *10 westpark mintlaw, AB425HY* Enter patients phone or 0 to cancel: *07368350709* Enter patients email or 0 to cancel: *jayden@gmail.com*	pass

	patients firstname: jayden patients surname: wilson patients address: 10 westpark mintlaw, AB425HY patients phone: 07368350709 patients email: jayden@gmail.com				
Update patient	Patient ID: patient600 Include insurance? [yes] [no]: no Include primary doctor? [yes] [no]: yes Doctor ID: 3527 patients firstname: jayden-james Patients surname: wilson patients address: address patients phone: phone number patients email: email	Should update patient	Does update patient	 Email should check for “@” symbol but does not impact functionality	pass
Delete patient	Patient id: patient600	Should delete patient	Does delete patient		pass

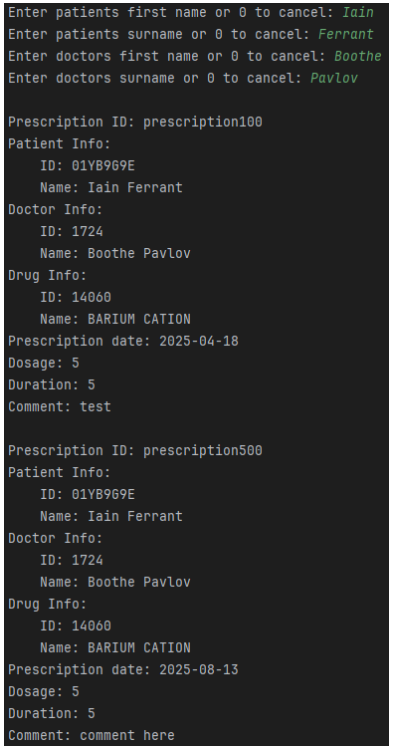
View all patients	Limit: 2	Should show all patient information within the limit	Does show all patient details within the limit		pass
View patient by id	Patientid: PHXNAQJ0	Should show details of patient by id	Does show details of patient by id		pass

View patient by first name and surname	Patient firstname: Janelle Patient surname: Behnecken	Shows a list of patients that match the same first name and surname	Shows a list of patients that match the same first name and surname		pass
Add doctor	Is the doctor a specialist? [yes] [no]: no DoctorId: doctor400 doctors firstname: jenny doctors surname: smith doctors address: 10 westpakr mintlaw ab425hy doctors email: jenny@outlook.com doctors hospital: mintlaw	Adds a new doctor	Adds a new doctor		pass
Update doctor	Is the doctor a specialist? [yes] [no]: no Doctor Id: doctor400 doctors firstname: jennyanne doctors surname: smith doctors address: 10 westpark mintlaw doctors email: jenny@outlook.com doctors hospital: mintlaw	Updates doctor	Updates doctor		pass

Delete doctor	doctorId: doctor400	Deletes doctor	Deletes doctor	Enter an option (0-7): 3 Enter Doctor ID or 0 to cancel: doctor400	pass
View doctor by id	doctorId: 3326	View doctor details by id	View doctor details by id	Enter Doctor ID or 0 to cancel: 3326 Doctor ID: 3326 Doctor firstname: Appolonia Doctor surname: Streatfeild Doctor address: PO Box 77011 Doctor email: astreatfeild7@jugem.jp Situated hospital: City Centre Specialization: Oncologists Experience: 0	pass
View doctor by first name and surname	doctors firstname: jenny doctors surname: smith	View doctor details by first name and surname	View doctor details by first name and surname	Enter doctors firstname or 0 to cancel: jenny Enter doctors surname or 0 to cancel: smith Doctor ID: doctor400 Doctor firstname: jenny Doctor surname: smith Doctor address: 10 westpakr mintlaw ab425hy Doctor email: jenny@outlook.com Situated hospital: mintlaw	pass
View doctors by speciality	speciality: 2 (NEUROLOGY)	View doctors by speciality	View doctors by speciality	Specialization Options 1: CARDIOLOGY 2: NEUROLOGY 3: PEDIATRICS 4: ONCOLOGY 5: ORTHOPEDICS 6: PSYCHIATRY Enter doctors specialization (1-6) or 0 to cancel: 2 Doctor ID: doc10 Doctor firstname: jayden Doctor surname: wilson Doctor address: address Doctor email: email Situated hospital: hospital Specialization: NEUROLOGY Experience: 10	pass

View all doctors	Limit: 2	View details of all doctors	View details of all doctors		pass
Add prescription	Prescription ID: prescription100 Patient ID: 01YB9G9E Drug ID: 14060 Doctor ID: 1724 prescription date (yyyy-mm-dd) : 2025-04-18 dosage: 5 duration: 3 comment: test	Add a prescription	Fails to add prescription as drug id does not exist. Cause: Constructor parameter order when creating a new prescription was in the wrong order. But is now fixed		Fail

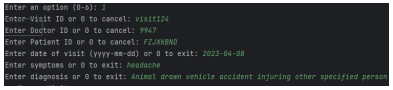
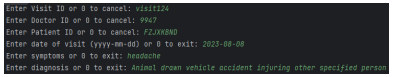
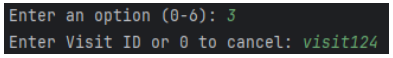
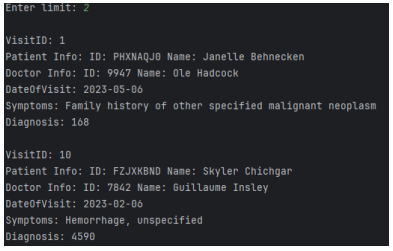
Update prescription	Prescription ID: prescription100 Patient ID: 01YB9G9E Drug ID: 15493 Doctor ID: 1724 prescription date (yyyy-mm-dd) : 2025-04-18 dosage: 5 duration: 3 comment: test	Update prescription	Update prescription	Enter an option (0-6): 2 Enter Prescription ID or 0 to cancel: prescription100 Enter Patient ID or 0 to cancel: 01YB9G9E Enter Drug ID or 0 to cancel: 1026382 Conflict found: BARIUM CATION (14060) Enter Drug ID or 0 to cancel: 15493 Enter Doctor ID or 0 to cancel: 1724 Enter prescription date (yyyy-mm-dd) or 0 to exit: 2025-04-04 Enter dosage or 0 to exit: 5 Enter duration or 0 to exit: 5 Enter comment or 0 to exit: 5 pills a week for 5 weeks	pass
Delete prescription	Prescription ID: prescription100	Delete prescription	Delete prescription	Enter an option (0-6): 3 Enter Prescription ID or 0 to cancel: prescription100	pass
View prescription by id	Prescription ID: prescription100	View prescription details by id	View prescription details by id	Enter Prescription ID or 0 to cancel: prescription100 Prescription ID: prescription100 Patient Info: ID: 01YB9G9E Name: Iain Ferrant Doctor Info: ID: 1724 Name: Boothe Pavlov Drug Info: ID: 14060 Name: BARIUM CATION Prescription date: 2025-04-18 Dosage: 5 Duration: 5 Comment: test	pass

View prescription by doctor and patient names	patients first name: Iain patients surname: Ferrant doctors first name: Boothe doctors surname: Pavlov	Display prescription details by patient and doctors names	Display prescription details by patient and doctors names		pass

View all prescriptions	Limit: 2	View details of all prescriptions within limits	View details of all prescriptions within limits	Enter limit: 2 Prescription ID: 1383892245 Patient Info: ID: FZJXKBND Name: Skyler Chichgar Doctor Info: ID: 9947 Name: Ole Hadcock Drug Info: ID: 34154 Name: risperidone Prescription date: 2023-04-08 Dosage: 16 Duration: 98 Comment: Animal-drawn vehicle accident injuring other specified person Prescription ID: 1634654226 Patient Info: ID: WSA59Z49 Name: Clevie Billsberry Doctor Info: ID: 4449 Name: Romola Thurber Drug Info: ID: 23810 Name: Aspen Prescription date: 2023-08-02 Dosage: 24 Duration: 34 Comment: Accidental poisoning by arsenic and its compounds and fumes	pass
Conflict during prescription management	Drug ID: 1026382	Display conflicts if found	Display conflicts if found. Conflict found: BARIUM CATION (14060)	Enter an option (0-6): 1 Enter Prescription ID or 0 to cancel: pre150 Enter Patient ID or 0 to cancel: 01YB9G9E Enter Drug ID or 0 to cancel: 1026382 Conflict found: BARIUM CATION (14060) Enter Drug ID or 0 to cancel:	pass
Add drug	Drug ID: drug315 drug name: moscovium side effects: death benefits: none to be found	Add drug	Add drug	Enter Drug ID or 0 to cancel: drug315 Enter drug name or 0 to exit: moscovium Enter side effects or 0 to cancel: death Enter benefits or 0 to cancel: none to be found	pass
Update drug	Drug ID: drug315 drug name: moscovium side effects: death benefits: none	Update drug	Update drug	Enter an option (0-6): 2 Enter Drug ID or 0 to cancel: drug315 Enter drug name or 0 to exit: moscovium Enter side effects or 0 to cancel: death Enter benefits or 0 to cancel: none	pass
Delete drug that is part of a prescription	Drugid: 14060	As the drug is part of existing prescriptions	As the drug is part of existing prescriptions its	Enter an option (0-6): 3 Enter Drug ID or 0 to cancel: 14060 (conn=347) Cannot delete or update a parent row: a foreign key constraint fails Need to add proper errors for	pass

		its prevented from being deleted	prevented from being deleted	foreign key errors	
View drug by id	Drugid: drug315	View details of drug by id	View details of drug by id	Enter Drug ID or 0 to cancel: drug315 Drug id: drug315 Drug name: moscovium Side effects: death Benefits: none	pass
View all drugs	limit : 2	View details of drugs within limit	View details of drugs within limit	Enter limit: 2 Drug id: 1026382 Drug name: test Side effects: test Benefits: test Drug id: 10987654 Drug name: test Side effects: test Benefits: test	pass
View Drug by drug name	Drug name: moscovium	View details of drug by drug name	View details of drug by drug name	Enter drugs name or 0 to cancel: moscovium Drug id: drug315 Drug name: moscovium Side effects: death Benefits: none	pass
Delete drug	Drugid: drug315	Delete drug	Delete drug	Enter an option (0-6): 3 Enter Drug ID or 0 to cancel: drug315	pass
Add insurance	Insurance Id: insurance145 insurance company name: HealthCo insurance address: 42 north	Add insurance	Add insurance	Enter Insurance ID or 0 to cancel: insurance145 Enter insurance company name or 0 to cancel: HealthCo Enter insurance address or 0 to cancel: 42 north avenue abereen AB436HW Enter insurance phone or 0 to cancel: 345678765	pass

	avenue abereen AB436HW insurance phone: 345678765				
Update insurance	Insurance Id: insurance145 insurance company name: HealthCo insurance address: 42 north avenue aberdeen AB436HW insurance phone: 345678765	Update insurance	Update insurance	Enter Insurance ID or 0 to cancel: insurance145 Enter insurance company name or 0 to cancel: HealthCo Enter insurance address or 0 to cancel: 42 north avenue aberdeen AB436HW Enter insurance phone or 0 to cancel: 345678765	pass
Delete insurance	Insurance Id: insurance145	Delete insurance	Delete insurance	Enter an option (0-6): 3 Enter Insurance ID or 0 to cancel: insurance145	pass
View all insurance	Limit: 3	View details of insurance within limit	View details of insurance within limit	Enter load limit: 3 Insurance ID: 54ARPD9 Company name: Cantrell Drug Company Address: 17668 Nova Pass Phone: 964-658-4951 Insurance ID: 9BFYJXV Company name: Carilion Materials Management Address: 068 Hollow Ridge Parkway Phone: 478-698-2522 Insurance ID: BS03EM0 Company name: Bay State Medical Address: 61735 Northfield Circle Phone: 212-489-0603	pass
View insurance by id	Insurance Id: insurance145	View details of insurance by id	View details of insurance by id	Enter Insurance ID or 0 to cancel: insurance145 Insurance ID: insurance145 Company name: HealthCo Address: 42 north avenue aberdeen AB436HW Phone: 345678765	pass
View insurance by company name	insurance company name: healthCo	View details of insurance by company name	View details of insurance by company name	Enter insurance company name or 0 to cancel: healthCo Insurance ID: insurance145 Company name: HealthCo Address: 42 north avenue aberdeen AB436HW Phone: 345678765	pass

Add visit	Visit ID: visit124 Doctor Id: 9947 Patient ID: FZJXKBND date of visit (yyyy-mm-dd): 2023-04-08 symptoms: headache diagnosis: Animal drawn vehicle accident injuring other specified person	Add visit	Add visit		pass
Update visit	Visit ID: visit124 Doctor Id: 9947 Patient ID: FZJXKBND date of visit (yyyy-mm-dd): 2023-08-08 symptoms: headache diagnosis: Animal drawn vehicle accident injuring other specified person	Update visit	Update visit		pass
Delete visit	Visit ID: visit124	Delete visit	Delete visit		pass
View all visits	Limit: 2	View details of visits within limit	View details of visits within limit		pass

View visits by id	Visit ID: visit124	View details of visit by id	View details of visit by id	Enter Visit ID or 0 to cancel: visit124 VisitID: visit124 Patient Info: ID: FZJXKBND Name: Skyler Chichgar Doctor Info: ID: 9947 Name: Ole Hadcock DateOfVisit: 2023-08-08 Symptoms: headache Diagnosis: Animal drawn vehicle accident injuring other specified person	pass
View visits by patient and doctors name	patients first name: Skyler patients surname: Chichgar doctors first name: Ole doctors surname: Hadcock	View visit details by patient and doctors name	View visit details by patient and doctors name	VisitID: visit124 Patient Info: ID: FZJXKBND Name: Skyler Chichgar Doctor Info: ID: 9947 Name: Ole Hadcock DateOfVisit: 2023-08-08 Symptoms: headache Diagnosis: Animal drawn vehicle accident injuring other specified person	pass
Add conflict	Drug ID: 55269 ID of conflicting Drug: 99089	Add conflict	Add conflict	Enter Drug ID or 0 to cancel: 55269 Enter ID of conflicting Drug or 0 to finish: 99089	pass
View Conflicts by Drug id	Drug ID: 55269	View conflicts details by Drug id	View conflicts details by Drug id	Enter Drug ID or 0 to cancel: 55269 Conflicting drugs ACTIVATED CHARCOAL (88506) SODIUM BICARBONATE (99089)	pass
Delete conflict	Drug I: 55269 ID of conflicting: 88506	Delete conflict	Delete conflict	Enter Drug ID or 0 to cancel: 55269 Enter ID of conflicting Drug or 0 to finish: 88506	pass